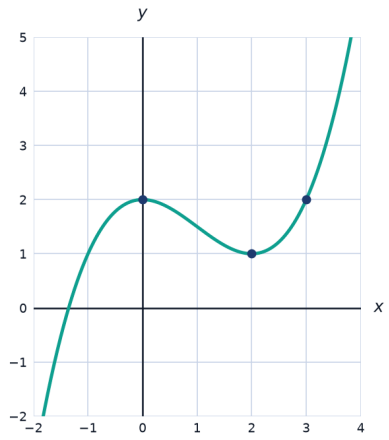
Proofs and further applications of polynomial identities

Core Concepts: Proofs and further applications of polynomial identities

1 For the model $f(x) = 4x + 9$ in proofs and further applications of polynomial identities, evaluate $f(2)$.



a. 17

b. 21

c. 8

d. 21

2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

3 A data set recorded for proofs and further applications of polynomial identities consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 4x + 9$ in proofs and further applications of polynomial identities, evaluate $f(3)$.

a. 21

b. 25

c. 12

d. 25

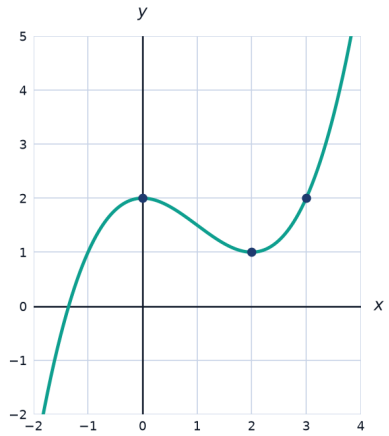
5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$



6 A data set recorded for proofs and further applications of polynomial identities consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 4x + 9$ in proofs and further applications of polynomial identities, evaluate $f(4)$.

a. 25

b. 29

c. 16

d. 29

8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

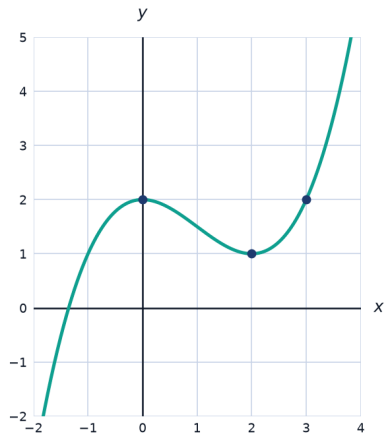
9 A data set recorded for proofs and further applications of polynomial identities consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

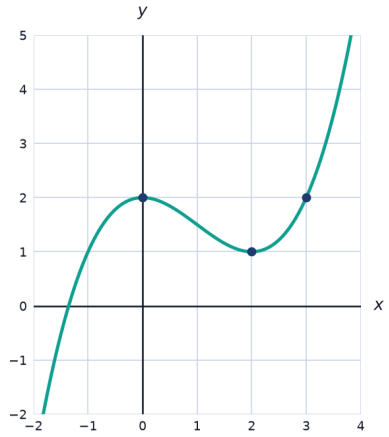
d. 18.5



Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 4x + 9$ in proofs and further applications of polynomial identities, evaluate $f(5)$.



a. 29

b. 33

c. 20

d. 33

11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

12 A data set recorded for proofs and further applications of polynomial identities consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

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